

Tutorial: Determinants

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Introduction

This tutorial provides carefully selected practice problems with detailed, step-by-step solutions for Chapter 3: Determinants. The problems are organized by topic and progressively build on the concepts covered in the lectures. Each solution emphasizes the reasoning behind every step so that you can develop a strong conceptual foundation alongside computational fluency.

Topic 1: Determinant of a 2×2 Matrix and Geometric Interpretation

Problem 1. Let $A = \begin{bmatrix} -2 & 5 \\ 3 & -4 \end{bmatrix}$.

- (a) Compute $\det(A)$.
- (b) Find the area of the parallelogram determined by the column vectors of A .
- (c) Does the matrix reverse or preserve the standard orientation of $\mathbb{R}^2$?

Solution.

(a) Using the 2×2 determinant formula $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$:

$$\begin{aligned}\det(A) &= (-2)(-4) - (5)(3) \\ &= 8 - 15 \\ &= -7\end{aligned}$$

(b) The geometric interpretation states that for a 2×2 matrix, the absolute value of the determinant equals the area of the parallelogram determined by its column vectors. Therefore:

$$\text{Area} = |\det(A)| = |-7| = 7$$

(c) Since $\det(A) = -7 < 0$, the linear transformation defined by A **reverses** the standard orientation of $\mathbb{R}^2$. The column vectors $\mathbf{a}_1 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 5 \\ -4 \end{bmatrix}$ form a left-handed (clockwise) ordered pair.

Answer: (a) $\det(A) = -7$ (b) Area = 7 (c) Reverses orientation

Topic 2: Cofactor Expansion — Strategic Choice of Row/Column

Problem 2. Compute $\det(A)$ for

$$A = \begin{bmatrix} 0 & 3 & -1 \\ 2 & 0 & 4 \\ -5 & 1 & 0 \end{bmatrix}$$

by choosing the most efficient row or column for cofactor expansion.

Solution. We inspect all rows and columns to find the one with the most zeros:

- Row 1: one zero ✓
- Column 1: one zero ✓
- Row 2: one zero
- Column 2: one zero
- Row 3: one zero
- Column 3: one zero

Several options are tied. We choose to expand along **column 1** (it has a zero in position $(1, 1)$, which eliminates one term):

$$\det(A) = a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} = 0 \cdot C_{11} + 2 \cdot C_{21} + (-5) \cdot C_{31}$$

Since the first term is zero, we only need two cofactors.

Computing C_{21} :

$$\begin{aligned} C_{21} &= (-1)^{2+1} \det(A_{21}) = (-1)^3 \det \begin{bmatrix} 3 & -1 \\ 1 & 0 \end{bmatrix} \\ &= -[(3)(0) - (-1)(1)] = -(0 + 1) = -1 \end{aligned}$$

Computing C_{31} :

$$\begin{aligned} C_{31} &= (-1)^{3+1} \det(A_{31}) = (-1)^4 \det \begin{bmatrix} 3 & -1 \\ 0 & 4 \end{bmatrix} \\ &= +[(3)(4) - (-1)(0)] = 12 - 0 = 12 \end{aligned}$$

Combining:

$$\det(A) = 2(-1) + (-5)(12) = -2 - 60 = -62$$

Answer: $\det(A) = -62$

Topic 3: Cofactor Expansion for a 4×4 Matrix

Problem 3. Compute $\det(A)$ for the 4×4 matrix

$$A = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 2 & -1 & 4 \\ 0 & 1 & 5 & -3 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Solution. We observe that **column 1** contains three zeros. Expanding along column 1:

$$\det(A) = a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} + a_{41}C_{41} = 1 \cdot C_{11} + 0 + 0 + 0$$

So we only need C_{11} :

$$C_{11} = (-1)^{1+1} \det(A_{11}) = \det \begin{bmatrix} 2 & -1 & 4 \\ 1 & 5 & -3 \\ 0 & 1 & 1 \end{bmatrix}$$

Now we compute this 3×3 determinant. Column 1 of this submatrix has a zero, so we expand along it:

$$\begin{aligned} \det \begin{bmatrix} 2 & -1 & 4 \\ 1 & 5 & -3 \\ 0 & 1 & 1 \end{bmatrix} &= 2 \cdot \det \begin{bmatrix} 5 & -3 \\ 1 & 1 \end{bmatrix} - 1 \cdot \det \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix} + 0 \cdot (\dots) \\ &= 2[(5)(1) - (-3)(1)] - 1[(-1)(1) - (4)(1)] \\ &= 2(5 + 3) - 1(-1 - 4) \\ &= 2(8) - 1(-5) \\ &= 16 + 5 = 21 \end{aligned}$$

Therefore:

$$\det(A) = C_{11} = 21$$

Answer: $\det(A) = 21$

Topic 4: Computing Determinants Using Row Operations

Problem 4. Use row operations to compute $\det(A)$ for

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 1 & -1 \\ 2 & 5 & -5 \end{bmatrix}$$

Solution. The strategy is to create zeros below the pivot in column 1 using Type I row operations (which do not change the determinant).

Step 1: $R_2 \rightarrow R_2 - 4R_1$ and $R_3 \rightarrow R_3 - 2R_1$:

$$\begin{bmatrix} 1 & -2 & 3 \\ 4 & 1 & -1 \\ 2 & 5 & -5 \end{bmatrix} \xrightarrow[R_3-2R_1]{R_2-4R_1} \begin{bmatrix} 1 & -2 & 3 \\ 0 & 9 & -13 \\ 0 & 9 & -11 \end{bmatrix}$$

Both operations are Type I (adding a multiple of one row to another), so $\det$ is unchanged.

Step 2: Create a zero in position $(3, 2)$ using $R_3 \rightarrow R_3 - R_2$:

$$\xrightarrow{R_3-R_2} \begin{bmatrix} 1 & -2 & 3 \\ 0 & 9 & -13 \\ 0 & 0 & 2 \end{bmatrix}$$

This is also a Type I operation, so the determinant is still unchanged.

Step 3: The matrix is now upper triangular. The determinant is the product of the diagonal entries:

$$\det(A) = 1 \times 9 \times 2 = 18$$

Answer: $\det(A) = 18$

Topic 5: Properties of Determinants — Product, Scalar, Transpose

Problem 5. Let A and B be 3×3 matrices with $\det(A) = 3$ and $\det(B) = -2$. Compute each of the following:

- (a) $\det(AB)$
- (b) $\det(2A)$
- (c) $\det(A^T B)$
- (d) $\det(A^{-1} B)$
- (e) $\det(2A^3 B^T)$

Solution. We apply the key properties systematically. Recall: A and B are 3×3 , so $n = 3$.

(a) Product rule: $\det(AB) = \det(A) \det(B)$:

$$\det(AB) = (3)(-2) = -6$$

(b) Scalar rule: $\det(kA) = k^n \det(A)$ with $n = 3$:

$$\det(2A) = 2^3 \cdot \det(A) = 8 \times 3 = 24$$

(c) Transpose rule: $\det(A^T) = \det(A)$, then product rule:

$$\det(A^T B) = \det(A^T) \det(B) = \det(A) \det(B) = (3)(-2) = -6$$

(d) Inverse rule: $\det(A^{-1}) = 1/\det(A)$, then product rule:

$$\det(A^{-1} B) = \det(A^{-1}) \det(B) = \frac{1}{3} \times (-2) = -\frac{2}{3}$$

(e) Combining all rules — scalar, power, transpose, product:

$$\begin{aligned} \det(2A^3 B^T) &= \det(2A^3) \cdot \det(B^T) && \text{(product rule)} \\ &= 2^3 \det(A^3) \cdot \det(B) && \text{(scalar rule with } n = 3; \text{ transpose rule)} \\ &= 8 \cdot [\det(A)]^3 \cdot (-2) && (\det(A^k) = [\det(A)]^k) \\ &= 8 \times 27 \times (-2) = -432 \end{aligned}$$

Answer: (a) -6 (b) 24 (c) -6 (d) $-\frac{2}{3}$ (e) -432

Topic 6: Determinants and Invertibility — Parameter Problems

Problem 6. Find all values of λ for which the matrix

$$A = \begin{bmatrix} \lambda & 1 & 0 \\ 1 & \lambda & 1 \\ 0 & 1 & \lambda \end{bmatrix}$$

is singular (i.e., non-invertible). Justify your answer using the Invertible Matrix Theorem.

Solution. By the Invertible Matrix Theorem, A is singular if and only if $\det(A) = 0$. We compute $\det(A)$ by expanding along **row 1** (which contains a zero):

$$\begin{aligned} \det(A) &= \lambda \cdot \det \begin{bmatrix} \lambda & 1 \\ 1 & \lambda \end{bmatrix} - 1 \cdot \det \begin{bmatrix} 1 & 1 \\ 0 & \lambda \end{bmatrix} + 0 \cdot (\dots) \\ &= \lambda(\lambda^2 - 1) - 1(\lambda - 0) \\ &= \lambda^3 - \lambda - \lambda \\ &= \lambda^3 - 2\lambda \\ &= \lambda(\lambda^2 - 2) \end{aligned}$$

Setting $\det(A) = 0$:

$$\lambda(\lambda^2 - 2) = 0 \implies \lambda = 0 \text{ or } \lambda^2 = 2 \implies \lambda = 0, \sqrt{2}, -\sqrt{2}$$

Conclusion: A is singular for $\lambda \in \{0, \sqrt{2}, -\sqrt{2}\}$, and invertible for all other values of λ .

Answer: $\lambda = 0, \sqrt{2}, -\sqrt{2}$

Topic 7: Cramer's Rule

Problem 7. Use Cramer's Rule to solve the system:

$$\begin{cases} x + 2y + z = 4 \\ 3x - y + 2z = 7 \\ 2x + y - z = 1 \end{cases}$$

Solution. Write the system as $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & -1 & 2 \\ 2 & 1 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 7 \\ 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Step 1: Verify Cramer's Rule applies by checking $\det(A) \neq 0$. Expand along row 1:

$$\begin{aligned} \det(A) &= 1 \cdot \det \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} - 2 \cdot \det \begin{bmatrix} 3 & 2 \\ 2 & -1 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix} \\ &= 1(1 - 2) - 2(-3 - 4) + 1(3 + 2) \\ &= -1 + 14 + 5 = 18 \neq 0 \quad \checkmark \end{aligned}$$

Step 2: Compute $x = \frac{\det(A_1(\mathbf{b}))}{\det(A)}$. Replace column 1 of A with $\mathbf{b}$:

$$\begin{aligned} \det(A_1(\mathbf{b})) &= \det \begin{bmatrix} 4 & 2 & 1 \\ 7 & -1 & 2 \\ 1 & 1 & -1 \end{bmatrix} \\ &= 4 \cdot \det \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} - 2 \cdot \det \begin{bmatrix} 7 & 2 \\ 1 & -1 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 7 & -1 \\ 1 & 1 \end{bmatrix} \\ &= 4(-1) - 2(-9) + 1(8) = -4 + 18 + 8 = 22 \\ x &= \frac{22}{18} = \frac{11}{9} \end{aligned}$$

Step 3: Compute $y = \frac{\det(A_2(\mathbf{b}))}{\det(A)}$. Replace column 2 of A with $\mathbf{b}$:

$$\begin{aligned} \det(A_2(\mathbf{b})) &= \det \begin{bmatrix} 1 & 4 & 1 \\ 3 & 7 & 2 \\ 2 & 1 & -1 \end{bmatrix} \\ &= 1 \cdot \det \begin{bmatrix} 7 & 2 \\ 1 & -1 \end{bmatrix} - 4 \cdot \det \begin{bmatrix} 3 & 2 \\ 2 & -1 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 3 & 7 \\ 2 & 1 \end{bmatrix} \\ &= 1(-9) - 4(-7) + 1(-11) = -9 + 28 - 11 = 8 \end{aligned}$$

$$y = \frac{8}{18} = \frac{4}{9}$$

Step 4: Compute $z = \frac{\det(A_3(\mathbf{b}))}{\det(A)}$. Replace column 3 of A with $\mathbf{b}$:

$$\begin{aligned}\det(A_3(\mathbf{b})) &= \det \begin{bmatrix} 1 & 2 & 4 \\ 3 & -1 & 7 \\ 2 & 1 & 1 \end{bmatrix} \\ &= 1 \cdot \det \begin{bmatrix} -1 & 7 \\ 1 & 1 \end{bmatrix} - 2 \cdot \det \begin{bmatrix} 3 & 7 \\ 2 & 1 \end{bmatrix} + 4 \cdot \det \begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix} \\ &= 1(-8) - 2(-11) + 4(5) = -8 + 22 + 20 = 34 \\ z &= \frac{34}{18} = \frac{17}{9}\end{aligned}$$

Verification: $x + 2y + z = \frac{11}{9} + \frac{8}{9} + \frac{17}{9} = \frac{36}{9} = 4 \quad \checkmark \quad 3x - y + 2z = \frac{33}{9} - \frac{4}{9} + \frac{34}{9} = \frac{63}{9} = 7$
 $\checkmark \quad 2x + y - z = \frac{22}{9} + \frac{4}{9} - \frac{17}{9} = \frac{9}{9} = 1 \quad \checkmark$

Answer: $x = \frac{11}{9}, \quad y = \frac{4}{9}, \quad z = \frac{17}{9}$

Topic 8: Adjugate Matrix and Formula for A^{-1}

Problem 8. Find A^{-1} using the adjugate formula $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$ for

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 1 \\ 0 & -1 & 2 \end{bmatrix}$$

Solution.

Step 1: Compute $\det(A)$. Expanding along row 1:

$$\begin{aligned}\det(A) &= 2 \cdot \det \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} - 1 \cdot \det \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} + 0 \cdot (\dots) \\ &= 2(6 + 1) - 1(-2 - 0) \\ &= 14 + 2 = 16\end{aligned}$$

Since $\det(A) = 16 \neq 0$, the inverse exists.

Step 2: Compute all nine cofactors.

Row 1:

$$C_{11} = (+1) \det \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = 7, \quad C_{12} = (-1) \det \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} = 2, \quad C_{13} = (+1) \det \begin{bmatrix} -1 & 3 \\ 0 & -1 \end{bmatrix} = 1$$

Row 2:

$$C_{21} = (-1) \det \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} = -2, \quad C_{22} = (+1) \det \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 4, \quad C_{23} = (-1) \det \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix} = 2$$

Row 3:

$$C_{31} = (+1) \det \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = 1, \quad C_{32} = (-1) \det \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} = -2, \quad C_{33} = (+1) \det \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = 7$$

Step 3: Form the cofactor matrix and its transpose (the adjugate).

$$C = \begin{bmatrix} 7 & 2 & 1 \\ -2 & 4 & 2 \\ 1 & -2 & 7 \end{bmatrix}, \quad \text{adj}(A) = C^T = \begin{bmatrix} 7 & -2 & 1 \\ 2 & 4 & -2 \\ 1 & -2 & 7 \end{bmatrix}$$

Step 4: Apply the formula.

$$A^{-1} = \frac{1}{16} \begin{bmatrix} 7 & -2 & 1 \\ 2 & 4 & -2 \\ 1 & -2 & 7 \end{bmatrix}$$

Topic 9: Determinants and Area in $\mathbb{R}^2$

Problem 9. Find the area of the triangle with vertices

$$P_1 = (1, 2), \quad P_2 = (4, 6), \quad P_3 = (5, 1).$$

Solution. The area of a triangle with vertices $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^2$ is given by:

$$\text{Area} = \frac{1}{2} \left| \det \begin{bmatrix} \mathbf{v}_2 - \mathbf{v}_1 & \mathbf{v}_3 - \mathbf{v}_1 \end{bmatrix} \right|$$

Step 1: Form the edge vectors from P_1 :

$$\mathbf{v}_2 - \mathbf{v}_1 = \begin{bmatrix} 4 - 1 \\ 6 - 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_3 - \mathbf{v}_1 = \begin{bmatrix} 5 - 1 \\ 1 - 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

Step 2: Compute the determinant:

$$\det \begin{bmatrix} 3 & 4 \\ 4 & -1 \end{bmatrix} = (3)(-1) - (4)(4) = -3 - 16 = -19$$

Step 3: Take half the absolute value:

$$\text{Area} = \frac{1}{2} |-19| = \frac{19}{2}$$

Answer: Area = $\frac{19}{2}$ square units

Topic 10: Determinants and Volume in $\mathbb{R}^3$

Problem 10. Find the volume of the parallelepiped determined by the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$$

Also determine whether $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ forms a right-handed or left-handed system.

Solution. The volume of the parallelepiped is $|\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]|$, and the sign of the determinant indicates orientation.

$$\det \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 0 \\ 3 & -1 & 4 \end{bmatrix}$$

Expanding along **row 2** (which contains a zero):

$$\begin{aligned} &= -(-2) \cdot \det \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix} - 0 \cdot (\dots) \\ &= 2(8 + 3) + 1(4 - 9) \\ &= 22 - 5 = 17 \end{aligned}$$

Volume:

$$\text{Volume} = |\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]| = |17| = 17$$

Orientation: Since $\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}] = 17 > 0$, the ordered set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ forms a **right-handed** system.

Answer: Volume = 17 cubic units; right-handed system

Topic 11: Linear Transformations and Volume Scaling

Problem 11. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation whose standard matrix is

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 1 \\ 0 & -2 & 4 \end{bmatrix}$$

A region $S \subset \mathbb{R}^3$ has volume 5. Find the volume of $T(S)$. Does T expand, contract, or preserve volume?

Solution. The volume scaling theorem states:

$$\text{Volume}(T(S)) = |\det(A)| \cdot \text{Volume}(S)$$

Step 1: Compute $\det(A)$ by expanding along row 1 (which has a zero):

$$\begin{aligned} \det(A) &= 2 \cdot \det \begin{bmatrix} 3 & 1 \\ -2 & 4 \end{bmatrix} - 1 \cdot \det \begin{bmatrix} -1 & 1 \\ 0 & 4 \end{bmatrix} + 0 \cdot (\dots) \\ &= 2(12 + 2) - 1(-4 - 0) \\ &= 28 + 4 = 32 \end{aligned}$$

Step 2: Apply the volume scaling formula:

$$\text{Volume}(T(S)) = |32| \times 5 = 160$$

Step 3: Since $|\det(A)| = 32 > 1$, the transformation T **expands** volume by a factor of 32. Every region in $\mathbb{R}^3$ is mapped to a region whose volume is 32 times the original.

Answer: $\text{Volume}(T(S)) = 160$; T expands volume by factor 32

Topic 12: Combined Properties Challenge

Problem 12. Let A be a 4×4 matrix with $\det(A) = \frac{1}{2}$. Compute each of the following *without* finding A explicitly, and state which property you use at each step:

- (a) $\det(3A)$
- (b) $\det(A^{-1})$
- (c) $\det(A^T)$
- (d) $\det(A^2)$
- (e) $\det(A^T A)$
- (f) $\det(2A^{-1}A^T)$

Solution. For a 4×4 matrix, $n = 4$. We apply the properties step by step.

(a) *Scalar multiplication rule:* $\det(kA) = k^n \det(A)$

$$\det(3A) = 3^4 \cdot \frac{1}{2} = 81 \times \frac{1}{2} = \frac{81}{2}$$

(b) *Inverse rule:* $\det(A^{-1}) = \frac{1}{\det(A)}$

$$\det(A^{-1}) = \frac{1}{1/2} = 2$$

(c) *Transpose rule:* $\det(A^T) = \det(A)$

$$\det(A^T) = \frac{1}{2}$$

(d) *Product rule:* $\det(A^2) = \det(AA) = \det(A) \det(A) = [\det(A)]^2$

$$\det(A^2) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

(e) *Transpose rule then product rule:*

$$\begin{aligned}\det(A^T A) &= \det(A^T) \cdot \det(A) = \det(A) \cdot \det(A) \\ &= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}\end{aligned}$$

(Note: $A^T A$ is always symmetric and its determinant is non-negative. Here $\frac{1}{4} \geq 0$, consistent.)

(f) *Combining scalar, inverse, transpose, and product rules:*

$$\begin{aligned}\det(2A^{-1}A^T) &= \det(2A^{-1}) \cdot \det(A^T) && \text{(product rule)} \\ &= 2^4 \det(A^{-1}) \cdot \det(A) && \text{(scalar rule; transpose rule)} \\ &= 16 \cdot \frac{1}{\det(A)} \cdot \det(A) && \text{(inverse rule)} \\ &= 16 \times 1 = 16\end{aligned}$$

(Insight: $A^{-1}A^T$ is an *orthogonal-like* matrix with $|\det| = 1$, and the scalar factor 2 contributes $2^4 = 16$.)

Answer: (a) $\frac{81}{2}$ (b) 2 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$ (e) $\frac{1}{4}$ (f) 16

Topic 13: Using Determinants to Test Linear Independence

Problem 13. Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$$

form a linearly independent set in $\mathbb{R}^3$. Use a determinant-based argument and verify your conclusion using the Invertible Matrix Theorem.

Solution. Three vectors in $\mathbb{R}^3$ are linearly independent if and only if the determinant of the matrix formed by placing them as columns is nonzero.

$$\det \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 2 \\ -2 & -1 & 5 \end{bmatrix}$$

Expanding along **column 1** (which has a zero):

$$\begin{aligned}&= 1 \cdot \det \begin{bmatrix} 2 & -1 \\ -1 & 5 \end{bmatrix} - 0 \cdot (\dots) + (-2) \cdot \det \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix} \\ &= 1(5 + 2) + (-2)(6 + 1) \\ &= 7 - 14 = -7\end{aligned}$$

Since $\det[\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3] = -7 \neq 0$, the vectors are **linearly independent**.

Verification via the Invertible Matrix Theorem: The matrix $M = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$ satisfies $\det(M) = -7 \neq 0$. By the Invertible Matrix Theorem, this is equivalent to each of the following:

- M is invertible ✓
- The columns of M (i.e., $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$) form a linearly independent set ✓
- The columns of M span $\mathbb{R}^3$ ✓
- $\text{rank}(M) = 3$ ✓
- The transformation $\mathbf{x} \mapsto M\mathbf{x}$ is one-to-one and onto ✓

Answer: Linearly independent (since $\det = -7 \neq 0$)